

Workbook: Body Cavities and Regions

Week 1 workbook · BIO 004 Human Anatomy · Summer 2026

A comprehensive workbook on body cavities, serous membranes, the mediastinum, the peritoneum, and the regional grids that anatomists and clinicians use to describe where things are. Work each question on your own first, then click Reveal. The structure of these questions tracks how cavities and regions are taught in any standard anatomy course.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Directional terms and body planes · 9 questions

Q01 · DOK 1 · Recall

Define anatomical position. Why does every directional term assume it?

Q02 · DOK 1 · Recall

Compare anterior and posterior, and give the synonyms used for each in human anatomy.

Q03 · DOK 1 · Recall

Compare medial and lateral. Give one example pair that uses each.

Q04 · DOK 1 · Recall

Compare superior and inferior. What synonym is sometimes used in place of each?

Q05 • DOK 2 • Apply

Proximal and distal apply primarily to the limbs. Define each, and explain why they are different from superior and inferior.

Q06 • DOK 1 • Recall

Compare superficial and deep. Give one example pair.

Q07 • DOK 1 • Recall

What is the midline of the body, and what is meant by ipsilateral versus contralateral?

Q08 • DOK 1 • Recall

Name the three primary anatomical planes and describe how each one cuts the body.

Q09 • DOK 1 • Recall

Distinguish midsagittal from parasagittal sections.

The body cavities • 8 questions

Q10 · DOK 1 · Recall

Name the two main body cavities and the subdivisions of each.

Q11 · DOK 1 · Recall

What is the cranial cavity, what surrounds it, and what does it contain?

Q12 · DOK 1 · Recall

What is the vertebral cavity, what surrounds it, and what does it contain?

Q13 · DOK 2 · Apply

Are the cranial and vertebral cavities continuous? How would you explain that connection structurally?

Q14 · DOK 1 · Recall

What does the thoracic cavity contain, and how is it subdivided?

Q15 · DOK 1 · Recall

What does the abdominopelvic cavity contain, and how is it subdivided?

Q16 • DOK 1 • Recall

What separates the thoracic cavity from the abdominopelvic cavity, and what kind of structure is it?

Q17 • DOK 1 • Recall

Name three smaller body cavities outside the dorsal and ventral cavities.

Serous membranes • 5 questions**Q18 • DOK 1 • Recall**

What is a serous membrane, what is it made of, and what does it do?

Q19 • DOK 1 • Recall

Compare the parietal and visceral layers of a serous membrane.

Q20 • DOK 1 • Recall

Name the three pairs of serous membranes in the body and where each pair sits.

Q21 • DOK 2 • Apply

Why does a serous cavity need fluid between the two layers, and what happens if too much accumulates?

Q22 • DOK 2 • Apply

Trace the pleura from the lung surface outward and name each part.

The mediastinum • 3 questions**Q23 • DOK 1 • Recall**

What is the mediastinum, what are its boundaries, and what does it contain?

Q24 • DOK 1 • Recall

Name the compartments of the mediastinum and what each contains.

Q25 • DOK 2 • Apply

Why is the mediastinum a high-stakes region for a chest wound or trauma?

The peritoneum · 6 questions

Q26 · DOK 1 · Recall

What is the peritoneum, and how do its two layers relate to abdominal organs?

Q27 · DOK 2 · Compare

Compare intraperitoneal and retroperitoneal organs by their relationship to the peritoneum.

Q28 · DOK 1 · Recall

What is a mesentery, what does it do, and give two examples.

Q29 · DOK 1 · Recall

What does the SADPUCKER mnemonic stand for? List the structures it names.

Q30 · DOK 1 · Recall

Name several intraperitoneal organs.

Q31 · DOK 3 · Reason

Peritonitis is inflammation of the peritoneum, often from a ruptured abdominal organ. Using what you know about the peritoneal cavity, predict why peritonitis can become a body-wide emergency so quickly.

Abdominopelvic regions (nine) · 5 questions

Q32 · DOK 1 · Recall

Why divide the abdomen into nine regions, and what kind of grid produces them?

Q33 · DOK 1 · Recall

Name the nine abdominopelvic regions in order from upper right across, middle row, and lower row.

Q34 · DOK 2 · Apply

Name one major organ found in each of the three top-row regions: right hypochondriac, epigastric, left hypochondriac.

Q35 • DOK 2 • Apply

Name one major organ found in each middle-row region: right lumbar, umbilical, left lumbar.

Q36 • DOK 2 • Apply

Name one major organ found in each bottom-row region: right iliac, hypogastric, left iliac.

Abdominopelvic quadrants (four) • 7 questions**Q37 • DOK 1 • Recall**

What two planes form the four abdominopelvic quadrants, and where do they cross?

Q38 • DOK 1 • Recall

Name the four abdominal quadrants.

Q39 • DOK 1 • Recall

Name the major organs found in the right upper quadrant.

Q40 • DOK 1 • Recall

Name the major organs found in the left upper quadrant.

Q41 • DOK 1 • Recall

Name the major organs found in the right lower quadrant.

Q42 • DOK 1 • Recall

Name the major organs found in the left lower quadrant.

Q43 • DOK 2 • Compare

Compare the quadrant system and the nine-region system. When is each one used?

Clinical reasoning • 4 questions**Q44 • DOK 3 • Reason**

A patient describes sharp pain in the right upper quadrant after a fatty meal. Using regional anatomy, list two organs you would consider first and explain why each fits the pain pattern.

Q45 · DOK 3 · Reason

A patient develops abdominal pain that starts around the umbilicus and over the next few hours migrates to the right lower quadrant. What is the most likely cause, and which abdominal region is involved?

Q46 · DOK 3 · Reason

A patient who fell on the left side of the abdomen develops left upper quadrant pain that also radiates to the left shoulder. Predict which organ has likely been injured and why the pain refers where it does.

Q47 · DOK 3 · Reason

A patient is stabbed in the chest. Air enters the pleural cavity. Predict what happens to the lung on that side and why.
